

A_∞ Enhancement of Geometry

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The goal of our research is to prove homological mirror symmetry conjecture or to develop the relevant area of study. The claim of homological mirror symmetry conjecture is A_∞ -equivalence of Fukaya category $Fuk(X)$ and A_∞ -enhancement $D^b(Coh(X))$ (typically noted as $D^b(X)$) of derived category of coherent sheaves. This statement says the elaboration of mirror symmetry, originally correspondence of symplectic data on Hodge structure and that of complex in Calabi-Yau manifold, whose symplectic side we name as A-model, the complex side as B-model, mirror symmetry solves many of QFT problem as well as Landau-Ginzburg model and others, and homological mirror symmetry is generalization of mirror symmetry in ∞ -category. The proof isn't easy, but the understanding of the problem statement itself is not difficult.

Definition 0.1. A_∞ Category

An A_∞ category $\mathcal{A}$ is a quasi-category which consists of objects $X \in Ob(\mathcal{A})$ and 1-morphisms $f \in Mor^1(\mathcal{A})$, and the 1-morphism contains higher coherence $\alpha : f \Rightarrow g$, called 2-morphism, and iteratively n -morphisms.

Definition 0.2. A_∞ Functor

Definition 0.3. Fukaya Category

For a symplectic manifold X , Fukaya category $Fuk(X)$ is an A_∞ category describing Lagrangian intersection. The objects of Fukaya category $Ob(Fuk(X))$ is Lagrangian submanifolds $L \subset X$, and the morphisms are Floer chain complex $CF^*(L_0, L_1)$ for each L_0 and L_1 . This $CF^*(L_0, L_1)$ is a $\mathbb{Z}$ -graded vector space defined by $CF^*(L_0, L_1) = \bigoplus_{p \in L_0 \cap L_1} \mathbb{k}p$. Floer chain complex is generated by its intersection points, and this is equivalent with counting critical points of Morse theory.

In Fukaya category, there is a composition $m_k : CF^*(L_{k-1}, L_k) \otimes \cdots \otimes CF^*(L_1, L_2) \rightarrow CF^*(L_0, L_k)[2-k]$ satisfying Stasheff associahedron property.

A_∞ enhancement of dg-category doesn't have unique enhancement, but the uniqueness is guaranteed if dg-category is derived category $D^b(X)$.

Definition 0.4. A_∞ Derived Category

A_∞ enhancement of derived category $D^b(X)$ is an A_∞ category $\mathcal{A}$, whose homology is $H^*(\mathcal{A}(E, F)) = Ext^*(E, F)$, while its trivialization $H^0(\mathcal{A}(E, F))$ needs to become the based derived category $H^0(\mathcal{A}(E, F)) \cong Hom_{D^b(X)}(E, F)$. This A_∞ enhancement $\mathcal{A}$ can be denoted as $D^b(X)$ by abuse of notation.

Conjecture 0.1. Homological Mirror Symmetry

There is an A_∞ equivalence of categories

$$Fuk(X) \cong D^b(X)$$

For solving or developing this statement, the natural question is if there is any geometric realization of A_∞ category, and another question is if there is any connection to the classical or derived algebraic geometry, and how that changes by deformation or quantization.

A_∞ algebra is a generalization of commutative ring. A_∞ category is its categorical generalization, especially if A_∞ category with one object is A_∞ algebra. I'll use A_∞ algebra to describe algebraic geometry. Recall a commutative ring defines an affine scheme by its ring spectrum $X = V(\mathfrak{p})$ with its structure sheaf $\mathcal{O}_X$, and that can also be described functorially as $F : CRing \rightarrow Set$. Hereafter I focus on functorial point of view.

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The proof of homological mirror symmetry conjecture is discussed from multi-dimensional perspective, including counting Gromow-Witten invariant, or Balmer spectrum, while generalizing formal geometry of non-unital A_∞ -algebra to A_∞ -category.

1 Derived Algebraic Geometry

2 Geometric Realization of A_∞

Definition 2.1. Counital Coalgebra

Coalgebra B is a graded vector space with morphism $\Delta : B \rightarrow B \otimes B$ s.t. $(\Delta \otimes id) \circ \Delta = (id \otimes \Delta) \circ \Delta$ called coproduct, and counital coalgebra is coalgebra B with unit $\epsilon : B \rightarrow 1$ where 1 is the unit object of $\mathcal{C}$, satisfying $(\epsilon \otimes id) \circ \Delta = (id \otimes \epsilon) \circ \Delta$. Counital coalgebra B is cofree counital coalgebra if it's given by $B = C(V) \cong T^c(V)$.

Definition 2.2. A Non-Commutative Thin Scheme

A non-commutative thin scheme X is a functor $X : Alg_{\mathcal{C}} \rightarrow Sets$, that is isomorphic to $X \cong Hom(-^*, B_X)$ for some coalgebra B_X , where $\mathcal{C}^f$ is artinian category of finite dimensional objects in $\mathcal{C}$, where $\mathcal{C}$ is $\mathbb{Z}$ -graded vector spaces. In particular, X is a non-commutative formal manifold, if the functor X is isomorphic to some $Spc(T(V))$ for a cofree counital tensor coalgebra $T(V) = \bigoplus_{n \geq 1} V^{\otimes n}$. The dimension of X is defined as $dim_k V$.

Definition 2.3. A Non-Commutative Thin DG-Scheme

A non-commutative thin dg-scheme is a pair (X, d) where X is a non-commutative thin scheme, and d is a vector field on X of degree +1 satisfying bracket of dgla as $[d, d] = 0$. This d is called homological vector field.

Definition 2.4. Non-Commutative Formal Pointed DG-Manifold

A non-commutative formal pointed dg-manifold is a pair $((X, x_0), d_X)$ such that (X, x_0) is a non-commutative formal pointed graded manifold, and d_X is a homological vector field on X such that $d_X|_{x_0} = 0$.

Proposition 2.1. Taylor Decomposition of Graded Vector Field

For the case of formal manifold X , let the corresponding cofree counital coalgebra be $B_X \cong T^c(V)$, and the homological vector field be coderivation $d_X : B_X \rightarrow B_X$, that can have restriction by cofreeness, thus

by using projection, $\pi \circ d_X : B_X \rightarrow V[1]$. If in addition, d_X can be decomposed degree-wise for its linearity, as $d_X = \sum_{n \geq 1} d_X^{(n)}$ for linear maps $d_X^{(n)} : V^{\otimes n} \rightarrow V[1]$.

Definition 2.5. Non-Unital A_∞ Algebra

A non-unital A_∞ -algebra $A = (A, m_n)$ is a graded vector space A and coherence condition $m_n : A^{\otimes n} \rightarrow A[2-n]$ with Stasheff relation, whose geometric interpretation is the structure of a nc fp dg-manifold $((X, x_0), d_X)$ together with a coalgebra isomorphism $B_X \cong T(T_{x_0}X)$ with $x_0 : k \rightarrow B_X$.

Definition 2.6. Structure of A_∞ Algebra

The structure of A_∞ Algebra on $V \in \text{Ob}(\text{Vect}_k^{\mathbb{Z}})$ is given by coderivation d of degree $+1$ of the non-counital cofree coalgebra $T_+(V[1]) = \bigoplus_{n \geq 1} V^{\otimes n}$ s.t. $[d, d] = 0$.

Proposition 2.2. Stasheff Relation

The condition of nc fp dg-manifold $((X, x_0), d_X)$ derives Stasheff relation of the A_∞ algebra, that is $m_n := d_X^{(n)}$ for $m_n : A^{\otimes n} \rightarrow A$ s.t. $\sum_{r+s+t=n} (-1)^{r+st} m_{r+1+t}(id^{\otimes r} \otimes m_s \otimes id^{\otimes t}) = 0$, whose degree is $|m_n| = 2 - n$. The Taylor decomposition of homological vector field $\pi \circ d_X = \sum_{n \geq 1} d_X^{(n)}$ is degree-wise by linearity, as $d_X^{(n)} : B_X^{(n)} \rightarrow B_X^{(1)}$.

Definition 2.7. Non-Commutative Formal n-Pointed DG-Manifold

Nc formal n -pointed dg-manifold $(\mathcal{X}, d_{\mathcal{X}}, \{pt_i\})$ is nc formal graded dg-manifold $(\mathcal{X}, d_{\mathcal{X}})$, and the points are $pt_i \in \mathcal{X}(k)$ for $1 \leq i \leq n$, and coderivation satisfies $d_{\mathcal{X}}|_{pt_i} = 0$, $d_{\mathcal{X}}|_{TX_i} : TX_i \rightarrow \bigoplus_{j,k} TX_j \oplus TX_k$ where X_i is formal neighborhood of X at pt_i .

Definition 2.8. A_∞ -Category

A_∞ -category $\mathcal{A}$ is nc formal dg-manifold $(\mathcal{X}, d_{\mathcal{X}}, \{pt_i\})$ with its coalgebra $B_{\mathcal{X}} \cong T^c(V[1])$. This V has a decomposition $V = \bigoplus_{i,j} E_{ij}$, where $E_{ij} := T(\widehat{X_i \times X_{j, pt_i, pt_j}})$.

Remark 2.1. T

is V is a k -linear associative algebra generated from k -linear category $\mathcal{C}$: let $\text{Ob}(\mathcal{C}) = I$ to define an algebra $V = \bigoplus \text{Hom}(i, j)$ for $i, j \in I$ is a k -linear algebra with the unit $1_V = \sum_{i \in I} id_i$, where $\pi_i = id_i \in V$ is a commuting idempotent i.e. $\pi_i^2 = \pi_i$, $\pi_i \pi_j = \pi_j \pi_i$. This makes homomorphism $k^I \rightarrow V$ of algebra with unit.

Proposition 2.3. Stasheff Relation

From non-unital A_∞ category $\mathcal{A}$, the coalgebra $d_{\mathcal{X}} : T^c(\bigoplus E_{ij}[1]) \rightarrow T^c(\bigoplus E_{ij}[1])$ satisfies the Stasheff relation. Consider the tensor coalgebra can be given by the grading structure $T^c(\bigoplus_{1 \leq i, j \leq n} E_{ij}[1]) = \bigoplus_{m \geq 0} \bigoplus_{1 \leq i_0, i_1, i_2, \dots, i_m \leq n} E_{i_0 i_1} \otimes E_{i_1 i_2} \otimes \dots \otimes E_{i_{m-1} i_m} := \bigoplus_{m \geq 0} B_{\mathcal{X}}^m$. From cofreeness of the coalgebra, the coderivation $d_{\mathcal{X}}$ canonically projected onto $\pi : B_{\mathcal{X}} \rightarrow B_{\mathcal{X}}^1$, and that can be decomposed linearly, so $\pi \circ d_{\mathcal{X}} = \sum d_{\mathcal{X}}^n$ where $d_{\mathcal{X}}^n : B_{\mathcal{X}}^n \rightarrow B_{\mathcal{X}}^1$, and we have Stasheff relation from $[d_{\mathcal{X}}, d_{\mathcal{X}}] = 0$, and $m_n := d_{\mathcal{X}}^n$.

Remark 2.2

$d_{\mathcal{X}} \circ \eta = 0$ implies $d_{\mathcal{X}} \circ \eta(1) = 0$ or $d_{\mathcal{X}}(1) = 0$. This means curvature is $m_0 = 0$.

Remark 2.3

For the case of $I = \{i\}$ consists of one vertex, we write $d|_{pt} = d \circ \eta$.

For the case of infinite objects A_∞ category is its inductive limit of finite objects.

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